Permutations & Combinations

The factorial function and counting arrangements

1 In how many ways can 6 different books be arranged in a row on a shelf?

2 In how many ways can 5 different keys be arranged in a row on a keyring holder?

3 In how many ways can 7 different trophies be arranged on a shelf?

4 In how many ways can 4 different paintings be hung in a row on a wall?

Permutations of a subset of items

5 In how many ways can 4 runners be chosen from a squad of 9 and assigned to first, second, third, and fourth positions in a relay?

6 In how many ways can 3 medals (gold, silver, bronze) be awarded to 3 of 8 competitors in a race?

7 In how many ways can 5 books be chosen from 12 and arranged in order on a shelf?

Combinations

8 A committee of 4 people is to be chosen from a group of 10. In how many ways can this be done?

9 A hand of 5 cards is dealt from a standard deck of 52 cards. In how many ways can this be done?

10 A pizza shop offers 8 toppings. In how many ways can a customer choose 3 toppings for a pizza?

11 A student must answer 5 out of 7 questions on an exam. In how many ways can the student choose which questions to answer?

Arrangements with repeated items

12 In how many distinct ways can the letters of the word "BANANA" be arranged?

13 In how many distinct ways can the letters of the word "MISSISSIPPI" be arranged?

14 In how many distinct ways can the letters of the word "STATISTICS" be arranged?

Arrangements with restrictions

15 In how many ways can 5 people be seated in a row if two particular people must sit next to each other?

16 In how many ways can 6 people be seated at a round table? (Rotations are considered identical arrangements.)

17 In how many ways can 7 people be seated in a row if two particular people must NOT sit next to each other?

18 In how many ways can 8 people be seated at a round table if two particular people must sit next to each other? (Rotations are considered identical arrangements.)

Combining permutations and combinations

19 A team of 3 is chosen from 5 boys and 4 girls, and must include exactly 2 boys and 1 girl. In how many ways can this be done?

20 A committee of 4 is chosen from 6 men and 3 women, and must include at least 1 woman. In how many ways can this be done?

21 A hand of 5 cards is dealt from a group of 4 aces and 8 other cards (12 cards total). Find the number of ways to choose a hand containing exactly 2 aces.

Using combinations in probability

22 A bag contains 6 red and 4 blue balls. Three balls are drawn at random without replacement. Find the probability that all three are red.

23 A bag contains 5 red and 5 blue balls. Two balls are drawn at random without replacement. Find the probability that both are blue.

24 A bag contains 7 red and 3 blue balls. Two balls are drawn at random without replacement. Find the probability that at least one is blue.

A well-designed word problem: forming a school quiz team

25 This question has two parts. A school has 12 eligible students (7 in Grade 11 and 5 in Grade 12) trying out for a quiz team of 4 students. (a) In how many ways can the team of 4 be chosen from all 12 students, with no restriction? (b) In how many ways can the team be chosen if it must contain exactly 2 students from each grade?